Problem Set for 28 April 2025

> **Important**
>
> The following exercises concern the theory of dual spaces and their associated structures.

Definition. (Kernels and Annulateurs). For convenience, let U_i denote subspaces of V, and B_i subspaces of V^*.

- $\operatorname{ann} U := \{f \in V^* \mid f(U) = 0\}$, a subspace of V^*;
- $\ker B := \{u \in V \mid f(u) = 0 \ (\forall f \in B)\} = \bigcap_{f \in B} \ker f$, a subspace of V.

These notions are defined for subsets in general. By the closure property:

- $\operatorname{ann}(X) = \operatorname{ann}(\operatorname{span}(X))$, for $X \subseteq V$;
- $\ker(S) = \ker(\operatorname{span}(S))$, for $S \subseteq V^*$.

Definition The 4 funcdamental spaces of a matrix

- $N(A)$: null space (column vectors),
- $C(A)$: column space (column vectors),
- $L(A)$: left null space (row vectors),
- $R(A)$: row space (row vectors).

Definition For $\varphi : U \to V$, we define the dual map

$$\varphi^* : V^* \to U^*, \quad \ell \mapsto \ell \circ \varphi.$$

In other word, φ^* means the pre-composition.

Exercise 0 (Very easy exercise). Show that φ^* is injective when φ is surjective.

> ⚠ **Warning**
>
> U and V are not finite dimensional in general, so never take the basis.

Exercise 1 Show that $\operatorname{ann}(\operatorname{im}\varphi)$ and $\ker(\varphi^*)$ are the same subspaces of V^*.

Tip

You are encouraged to restate the equality $\operatorname{ann}(\operatorname{im}\varphi) = \ker(\varphi^*)$ in natural language. One possible (oral) answer is:

- for any $f \in V^*$ killed by the $(- \circ \varphi)$, f kills all $\varphi(u)$'s.

Warning

U and V are not finite dimensional in general, so never take the basis.

Exercise 2 When φ is surjective, show that $\mathbf{im}(\varphi^*)$ and $\mathbf{ann}(\ker \varphi)$ are the same subspaces of U^*

> **Tip**
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> You are encouraged to restate the equality $\mathbf{im}(\varphi^*) = \mathbf{ann}(\ker \varphi)$ in natural language. One possible (oral) answer is:
>
> - for any $g \in U^*$ killing $\ker \varphi$, g factors through φ.

> **Warning**
>
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Exercise 3 When $\varphi : S \hookrightarrow V$ is injective, show that both $(V/S)^*$ and $\operatorname{ann}(S)$ are isomorphic to $\ker(\varphi^*)$.

> **Caution**
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> Can we write $(V/S)^* = \operatorname{ann}(S)$ here; consider the appropriate interpretation involving isomorphisms.

> **Warning**
>
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Exercise 4 For any V, we define the evaluation as before:

$$\Phi_V : V \to V^{**}, \quad v \mapsto \begin{bmatrix} V^* & \to & \mathbb{F} \\ \ell & \mapsto & \ell(v) \end{bmatrix}.$$

We define $\varphi^{**} := (\varphi^*)^*$, that is, the pre-composition of $\ell : U^* \to \mathbb{F}$ by $\varphi^* : V^* \to U^*$. Show the equality of the compositions

$$\left[U \xrightarrow{f} V \xrightarrow{\Phi_V} V^{**}\right] = \left[U \xrightarrow{\Phi_U} U^{**} \xrightarrow{f^{**}} V^{**}\right].$$

In other words, $\Phi_V(f(u)) = f^{**}(\Phi_U(u))$ for any $u \in U$. This is why we say Φ is natural.

> **Tip**
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> This can be demonstrated using string manipulations, even if the conceptual meaning is not yet fully clear.

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>
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Exercise 5 Show that

$$a : \mathbb{F}^{m\times n} \to (\mathbb{F}^{n\times m})^*, \quad M \mapsto \begin{bmatrix} \mathbb{F}^{n\times m} & \to & \mathbb{F} \\ X & \mapsto & \operatorname{trace}(M \cdot X) \end{bmatrix}$$

is a linear map. And show that a is surjective.

> **Tip**
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> Consider how to demonstrate the surjectivity efficiently?

Exercise 6 (For recreation). For finite dimensional vector space V over arbitrary field $\mathbb{F}$, we write

- $V := \mathbb{F}^{n\times 1}$, the space of columns vectors with common basis $\{e_i\}_{i=1}^n$, and
- $V^* := \mathbb{F}^{1\times n}$, the space of row vectors with common basis $\{f^j\}_{j=1}^n$,

Define a **non-degenerate bilinear pairing** via the standard row-column multiplication:

$$V^* \times V \to \mathbb{F}, \quad (u_{\text{row}}; v_{\text{column}}) \mapsto u \cdot v.$$

Using this pairing, observe:

1. The left multiplication of a matrix $(A\cdot) : V \to V$, induces a dual linear map of the right multiplication of the matrix $(\cdot A) : V^* \to V^*$. Remind the identity

$$(u; A\cdot v) = (u\cdot A; v).$$

2. Once V^* is viewed as a column space via the transpose $(-)^T : V^* \to V$, then:

$$(\cdot A) = V^* \xrightarrow{(-)^T} V \xrightarrow{A^T\cdot} V \xrightarrow{(-)^T} V^*.$$

   Hence, if $\varphi : U \to V$ has the matrix form $M \in \mathbb{F}^{m\times n}$, φ^* has the matrix form M^T. Now $(M^T)^T = M$ is consistent with the natural isomorphism.

The question: explain $\ker\varphi = \ker(\operatorname{im}(\varphi^*))$ by

$$N(A) = \ker(A\cdot) = \bigcap_{u\in V^*} \ker(uA\cdot) = \ker(\operatorname{im}((A\cdot)^*)) = (L(A^T))^T.$$

Analogously, explain $\mathrm{FD}\times\{\mathrm{Z}i\}_{i=1}^6$ in this exercise.